

Constrained Optimisation

LLE – Mathematics and Statistics Skills

1. You have the choice of two products, product x and product y . Product x has a cost of £4 and product y has a cost of £1. You have a maximum of £60 that can be used for purchases.

The amount of utility is given by the equation:

$$U(x, y) = x^2y$$

- (a) Maximum units of y with $x = 8$

$$8 \times £4 = £32$$

$$£60 - £32 = £28$$

$$£28 \div £1 = 28 \text{ items of } y$$

- (b) Utility for $(8, 28)$

$$U(8, 28) = 8^2 \times 28 = 1792$$

- (c) Utility for $x = 12$

$$12 \times £4 = £48$$

$$£60 - £48 = £12$$

$$£12 \div £1 = 12 \text{ items of } y.$$

$$U(12, 12) = 12^2 \times 12 = 1728$$

- (d) Constraint Function

$$4x + y \leq 60$$

- (e) Lagrangian Optimisation

The Lagrangian is $L(x, y, \lambda) = x^2y + \lambda(60 - 4x - y)$.

First-order conditions:

$$\frac{\partial L}{\partial x} = 2xy - 4\lambda = 0$$

$$\frac{\partial L}{\partial y} = x^2 - \lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = 60 - 4x - y = 0$$

The first two equations can be rearranged to:

$$2xy = 4\lambda$$

$$x^2 = \lambda$$

Dividing these equations gives

$$\frac{2y}{x} = 4 \implies y = 2x$$

Substitute this into the constraint [C]:

$$4x + 2x = 60 \implies x = 10.$$

Then $y = 20$.

$$U(10, 20) = 10^2 \times 20 = 2000$$

- (f) Since the utility from other points (1792, 1728) was lower than 2000, this is a maximum solution.

2.

$$B = r^2p^3$$

There is a maximum of 8 hours of time each day to prepare.

The constraint is $r + p \leq 8$.

The first-order conditions are:

$$\frac{\partial L}{\partial r} = 2rp^3 - \lambda = 0 \quad [A]$$

$$\frac{\partial L}{\partial p} = 3r^2p^2 - \lambda = 0 \quad [B]$$

From [A] and [B], we can establish the relationship $p = \frac{3}{2}r$.

Substitute this into the constraint:

$$r + \frac{3}{2}r = 8 \implies r = 3.2.$$

Then $p = 4.8$.

3. A utility function is given by $U(x, y) = x^{0.6}y^{0.4}$. The cost of good x is £5 and good y is £8. The budget is £1200.

The constraint is $5x + 8y \leq 1200$.

The Lagrangian is $L = x^{0.6}y^{0.4} + \lambda(1200 - 5x - 8y)$.

The first-order conditions are:

$$\frac{\partial L}{\partial x} = 0.6x^{-0.4}y^{0.4} - 5\lambda = 0$$

$$\frac{\partial L}{\partial y} = 0.4x^{0.6}y^{-0.6} - 8\lambda = 0$$

From these,

$$0.6x^{-0.4}y^{0.4} = 5\lambda$$

$$0.4x^{0.6}y^{-0.6} = 8\lambda$$

Dividing these two equations

$$\frac{3y}{2x} = \frac{5}{8} \implies y = \frac{5}{12}x$$

Substitute into the constraint:

$$5x + \frac{10}{3}x = 1200 \implies x = 144.$$

Then $y = 60$.

$$U = 144^{0.6} \times 60^{0.4} = 24.87$$

4. For the utility function of question 3, a utility of 75 must be achieved. The cost of goods x and y are £5 and £8.

(a) Cost Function

$$C = 5x + 8y$$

(b) Constraint Function**

$$x^{0.6}y^{0.4} = 75$$

(c) Lagrangian

We want to minimise cost, so the Lagrangian is:

$$L = 5x + 8y + \lambda(75 - x^{0.6}y^{0.4})$$

(d) Minimum Cost Solution

The first-order conditions are:

$$\frac{\partial L}{\partial x} = 5 - 0.6\lambda x^{-0.4}y^{0.4} = 0$$

$$\frac{\partial L}{\partial y} = 8 - 0.4\lambda x^{0.6}y^{-0.6} = 0$$

Rearranging these gives:

$$5 = 0.6\lambda x^{-0.4}y^{0.4}$$

$$8 = 0.4\lambda x^{0.6}y^{-0.6}$$

Dividing the two equations gives:

$$\frac{5}{8} = \frac{0.6\lambda x^{-0.4}y^{0.4}}{0.4\lambda x^{0.6}y^{-0.6}} = \frac{0.6}{0.4} \frac{y}{x} = \frac{3}{2} \frac{y}{x}$$

This simplifies to $10x = 24y$, or $x = \frac{12}{5}y$.

Substitute this into the constraint:

$$\left(\frac{12}{5}y\right)^{0.6}y^{0.4} = 75 \implies \left(\frac{12}{5}\right)^{0.6}y = 75$$

This leads to the integer solutions $y = 44$ and $x = 106$.

The cost is:

$$C = 5(106) + 8(44) = 530 + 352 = 882$$

5.

$$S = c^{0.25}f^{0.75}$$

It is important that my blood sugar is 100 units during each day.
The cost of 1 chocolate is 2p and the cost of 1 fudge is 8p.

We want to minimise cost $C = 2c + 8f$ subject to the constraint $c^{0.25}f^{0.75} = 100$.

The Lagrangian is $L(c, f, \lambda) = 2c + 8f + \lambda(100 - c^{0.25}f^{0.75})$.

The first-order conditions are:

$$\frac{\partial L}{\partial c} = 2 - 0.25\lambda c^{-0.75}f^{0.75} = 0$$

$$\frac{\partial L}{\partial f} = 8 - 0.75\lambda c^{0.25}f^{-0.25} = 0$$

$$\frac{\partial L}{\partial \lambda} = 100 - c^{0.25}f^{0.75} = 0$$

Rearranging the first two conditions and dividing gives:

$$\frac{2}{8} = \frac{0.25\lambda c^{-0.75}f^{0.75}}{0.75\lambda c^{0.25}f^{-0.25}} \implies \frac{1}{4} = \frac{1}{3} \frac{f}{c}$$

This simplifies to $3c = 4f$, or $c = \frac{4}{3}f$.

Substitute this into the constraint:

$$\left(\frac{4}{3}f\right)^{0.25}f^{0.75} = 100 \implies \left(\frac{4}{3}\right)^{0.25}f = 100$$

$$f = \frac{100}{(4/3)^{0.25}} \approx 93.06 \text{ and } c \approx 124.08$$

Since we must buy whole units, we test the integer point ($c=124$, $f=93$). It satisfies the constraint ($S \geq 100$) and is the cheapest valid option.

The new cost is:

$$C = 2(124) + 8(93) = 248 + 744 = 992$$